

AI LAB 99

Machine Learning Track — Week 4, Module 9

Deployment Readiness & Streamlit App Design Report

Reviewing the Full Project Pipeline and Planning the Customer Segmentation Dashboard

Dataset: Final Dataset.csv | Model: kmeans_model.pkl + standard_scaler.pkl

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Introduction

Before building the customer-segmentation Streamlit application, this module performs a final audit of everything produced across the project so far — the dataset, the engineered features, the trained model, the saved artifacts, and the resulting business segments — and then plans the dashboard's content, navigation, and technology stack.

Task 1 & 2 — Final Dataset Review

Final Dataset.csv was reloaded and confirmed as the single source of truth for the application: 2,038 rows and 42 columns, matching the output of Module 7/8 exactly (2,038 rows after the 202-duplicate removal documented in Module 5, 42 columns including the added Cluster and Customer_Segment labels).

1. Education	2. Marital_Status	3. Income	4. Recency
5. MntWines	6. MntFruits	7. MntMeatProducts	8. MntFishProducts
9. MntSweetProducts	10. MntGoldProds	11. NumDealsPurchases	12. NumWebPurchases
13. NumCatalogPurchases	14. NumStorePurchases	15. NumWebVisitsMonth	16. AcceptedCmp3
17. AcceptedCmp4	18. AcceptedCmp5	19. AcceptedCmp1	20. AcceptedCmp2
21. Complain	22. Response	23. Age	24. Year
25. Month	26. Day	27. Customer_Age	28. Customer_Tenure
29. Total_Children	30. Partner	31. Family_size	32. Total_spending
33. Total_Purchase	34. Total_Campaign	35. Avg Spending/Purchase	36. Digital_Engagement
37. Deal_Dependency	38. Preferred_Shopping_Channel	39. Product_Preference	40. Customer_Activity_Level
41. Cluster	42. Customer_Segment		

Note: as in Modules 7 and 8, every numeric column here is still standardized (StandardScaler) output, not real-world units. This matters directly for Task 8 below, where several dashboard panels are meant to show Income and spending figures to business users.

Task 3 — Final Engineered Features

All 13 features engineered in Module 5 were confirmed present in the final dataset:

Customer_Age	✓Present
Customer_Tenure	✓Present
Total_Children	✓Present
Family_size	✓Present
Total_spending	✓Present
Total_Purchase	✓Present

Total_Campaign	✓Present
Average Spending per Purchase	✓Present
Digital_Engagement	✓Present
Deal_Dependency	✓Present
Preferred_Shopping_Channel	✓Present
Product_Preference	✓Present
Customer_Activity_Level	✓Present

Task 4 — Final Customer Segmentation Model

Model	K-Means
Saved as	kmeans_model.pkl
Engineered feature dtypes	All 13 features are float64 — confirmed clustering-ready

Task 5 — Saved Model Files

A recursive search of the project directory located both artifacts the Streamlit app will need, saved together in the Module 6 folder:

kmeans_model.pkl	D:\AI-Lab-99\Week 4\Module 6 Customer Segmentation Model Development\
standard_scaler.pkl	D:\AI-Lab-99\Week 4\Module 6 Customer Segmentation Model Development\

This is good news for Task 8: because standard_scaler.pkl was saved alongside the model, the app can call scaler.inverse_transform() to convert Income, Total_spending, and similar columns back to real-world units before displaying them on any business-facing dashboard panel, rather than showing raw z-scores to end users.

Task 6 — Customer Segment Labels

Cluster 0	Premium Loyal Customers — 1,111 customers (54.5%)
Cluster 1	Digital-First Buyers — 927 customers (45.5%)

Naming has now changed twice for the same cluster: Cluster 1 was “Digital-First Buyers” in Module 7, “Budget Buyers” in Module 8, and is back to “Digital-First Buyers” here in Module 9 (because this notebook reloads Final Dataset.csv, which still carries the original Module 7 label rather than Module 8's renamed one). Before building the dashboard, pick one canonical name for each segment and use it consistently — whichever label the Streamlit app displays will become the name business stakeholders remember and use.

Task 7 — Business Segments

Two business segments were confirmed from the final labeled dataset: (1) Premium Loyal Customers, (2) Digital-First Buyers. As noted in the Module 7 and Module 8 reports, the 2-cluster model does not currently support the additional “At-Risk Customers” / “New Customers” / “Discount-Driven Families” segments referenced in earlier marketing materials — the dashboard should present only these two confirmed segments unless the model is retrained at a higher k.

Task 8 — Dashboard Information

The Streamlit dashboard is planned to display twelve items:

1. Total number of customers
2. Total number of customer segments
3. Customer distribution by segment
4. Income comparison
5. Total spending comparison
6. Product preferences
7. Shopping channel preferences
8. Customer activity levels
9. Campaign response
10. Customer persona
11. Customer segment prediction
12. Marketing recommendations

Build risk to flag now: items 4, 5, and 9 (Income comparison, Total spending comparison, Campaign response) will repeat the Module 8 dashboard bug — showing meaningless z-score totals like “-1.9” instead of dollar figures — unless the app explicitly inverse-transforms these columns with the saved `standard_scaler.pkl` before rendering them. Recall also that Campaign response (AcceptedCmp1–5, Response) was found to be entirely zero for both segments due to the Module 3 outlier-capping issue; that panel will show nothing meaningful until the upstream data-cleaning bug is fixed.

Task 9 — Dashboard Navigation

The application will use an eight-page navigation structure:

Home	Landing page / project summary
Dataset Overview	Shape, columns, and sample records
Customer Segments	Segment sizes and distribution
Business Dashboard	The 12 metrics/charts from Task 8
Customer Personas	Persona cards per segment
Predict Customer Segment	Live prediction on new customer input

Marketing Recommendations	Strategy tables from Module 8
About Project	Methodology and pipeline summary

Build risk to flag now: the “Predict Customer Segment” page will need to apply the exact same preprocessing chain used in training — missing-value handling, feature engineering, encoding, and StandardScaler transform (via `standard_scaler.pkl`) — to any new customer input before calling `kmeans_model.predict()`. Skipping any one of those steps will produce an invalid prediction. Reusing the `preprocessing_pipeline()` function documented in the Module 5 report is the safest way to guarantee this consistency.

Marketing Recommendations page caveat: the Module 8 report found that the merged strategy matrix intended to feed this page was empty (0 rows) due to a chained inner-join bug. That bug should be fixed before this page is wired up, or it will ship with no content.

Task 10 — Application Technology

Streamlit was selected as the application framework, for the following stated reasons:

- Python-based, so it integrates directly with the existing pandas/scikit-learn pipeline
- Easy integration with the trained machine learning model
- Supports Pandas and Plotly for tables and interactive charts
- Interactive user interface without needing separate frontend development
- Can load the saved K-Means model using Joblib
- Well suited to an ML internship-scale project

This is a reasonable, low-friction choice for a single-model, dashboard-style internal tool; no counter-arguments to this choice were raised in the notebook.

Deployment Readiness Summary

Dataset finalized	Yes — 2,038 × 42, all 13 engineered features present
Model finalized & saved	Yes — <code>kmeans_model.pkl</code> + <code>standard_scaler.pkl</code> located
Segment labels defined	Yes, but inconsistent across modules — needs a single canonical naming decision
Dashboard content planned	Yes — 12 items, 3 of which are currently at risk of showing invalid figures
Navigation planned	Yes — 8 pages, 2 of which depend on fixes from earlier modules
Technology selected	Yes — Streamlit, with clear justification

The project is close to deployment-ready. The three items below should be resolved before or during the build, in priority order:

- **1. Inverse-transform business-facing numbers:** wrap `Income`, `Total_spending`, and similar columns with `scaler.inverse_transform()` wherever they appear in the dashboard, not just internally.

- **2. Fix the Module 8 pipeline bugs before wiring the Marketing Recommendations page:** the strategy matrix feeding this page is currently empty.
- **3. Settle on one segment name per cluster:** “Digital-First Buyers” is recommended for consistency with Module 7 and this module, since it reflects the label actually stored in the final dataset.

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